MIL-A-36247A(DSA-DM)
18 June 1970
SUPERSEDING
MIL-A-36247
24 July 1964

MILITARY SPECIFICATION

AUDIOMETER, AUTOMATIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an automatic Bekesey-type audiometer, complete with mechanical recorder, earphones and a carrying case.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-M-7298	Manual, Technical: Commercial Equipment.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

NATIONAL BUREAU OF STANDARDS (NBS)

Close Coupler Type 9A

(Application for copies should be addressed to Superintendent of Documents, Washington, D. C. 20025.)

American National Standards Institute (ANSI)

S3.6 Specifications for Audiometers.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N. Y. 10018.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, suitable in all respects for the intended purpose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

* 3.2 Compliance with American National Standards Institute Specifications. Audiometer shall be in accordance with the requirements of ANSI S3.6. Compliance with these requirements does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his commodity. In the event of differences between the requirements of this specification and other specifications referenced herein, the requirements of this specification shall govern.

* 3.3 Style and design. Audiometer shall be a discrete frequency, automatic-recording, Bekesy-type, producing audio-frequency notes by means of a resistance-coupled oscillator. The audiometer shall consist essentially of the receiver, recorder, earphones, patient-control switch, audiogram charts, and carrying case. The audiometer shall have a zero reference level and shall be suitable for determining hearing loss in decibels. The audiometer shall be free from electrical-shock hazard when tested as specified in 4.4.7.

* 3.4 Attenuation. Audiometer shall possess an electrically-reversible, motor-driven attenuator unit controlled by a hand-operated switch. The maximum RMS voltage at the hand switch shall be 24 volts. The attenuator shall be of a ladder-type design having a nominal impedance of 100 ohms and a total attenuation range of 100 decibels (db). It shall be of a linear non-stepped cone structure, and that portion along which the moving contact slides shall be continuously wire-wound. For resolution accuracy, the design shall provide a minimum of 33 turns of wire between any two points corresponding to a separation of 10 db on the audiogram card. The maximum allowable tolerance shall be $\pm$ 1 db for any 10 db increment over the first 70 db of attenuation, but shall not exceed $\pm$ 3.5 db cumulative for the entire intensity range or any portion thereof. The system shall be calibrated so the total dynamic range of attenuation shall correspond to a Hearing Threshold Level Range on the audiogram card of 0 db normal to 100 db of loss in accordance with Table II of ANSI S3.6. The audiometer shall be tested as specified in 4.4.3 and 4.4.4 to determine compliance with accuracy requirements. Rate of change of attenuation shall be 5 db per second across the total dynamic range of 100 db.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

3.5 Frequency. The frequencies shall be fixed in discrete steps of 500, 1000, 2000, 3000, 4000 and 6000 cycles per second (cps). The audiometer shall present pure, nonpulsed tone. Frequency error shall not be greater than ± 3 percent for any given frequency. The accuracy of the frequencies shall be tested as specified in 4.4.1.

* 3.6 Earphones. The earphones shall be of the dynamic moving-coil type. The left and right earphones shall be marked with different colors or in another distinct manner for rapid identification. The earphones shall be individually calibrated at each of the six discrete frequencies. Calibration shall be held to ± 3 decibels for all frequencies with the exception that at 6000 cycles per second the tolerance shall be ± 5 decibels. The sound pressure level of the fundamental frequency shall be at least 30 decibels above that of any harmonic. The root mean square (rms) weighted sound pressure, produced by the earphones due to all frequencies except the signal frequency and its harmonics, shall be either less than 0.001 dynes per square centimeter or at least 60 decibels below the sound pressure, due to the signal frequency and its harmonics, at all designated frequencies and hearing loss levels. Any signal from the earphone not being used for test purposes shall have a sound pressure level at least 10 db below the hearing threshold reference level, except that it need not be more than 70 decibels below the signal from the active earphone. The headband shall exert a force of 1 to 2 kilograms laterally. The earphone bushing shall be constructed of sufficient strength to hold earphones properly positioned. Tests to determine conformance with the above requirements shall be conducted as specified in 4.4.2.

* 3.7 Patient control switch. Each audiometer shall be provided with a patient control switch, assembled to a rubber-insulated conductor at least 3 feet in length. The handswitch shall control the signal intensity across the hearing threshold so that the test tone will decrease when the button is depressed and increase when the button is released.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

* 3.8 Power supply. Audiometer shall be designed to operate selectively at 115 and 230 volts, 50 and 60 cycle, alternating current. Audiometer shall have an internal voltage regulation circuit to compensate for ± 10 percent fluctuations of line voltage while maintaining the required calibration accuracy. The acoustic output of the earphones shall not vary through a total range of more than 2 db as the line voltage is varied through the range required above.

3.9 Recording system. The audiometer shall have a mechanical recorder in a printing system using a typewriter ribbon. The recording stylus shall be positively synchronized with and securely engaged to the attenuator so that the hearing loss recorded on the audiogram card shall correspond to the attenuator setting within those limits of tolerance specified for the attenuator. The stylus shall press against the ribbon to make a dot on the audiogram card. The record shall be so controlled as to give 20 dots to the inch of vertical record. Ribbon shall have a minimum use life of 6 months. The ribbon shall be in two colors so that test and retest on the same card may be performed.

3.10 Performance. Audiometer shall be a completely automatic machine. Once the test has started, the machine shall continue until the test has been completed. At the end of the test, control shall be taken away from the subject and the signal shall be reduced to zero. The audiometer shall be designed so that the first test frequency of 500 cps shall be directed to the left earphone. The 500 cps signal shall be held for approximately 30 seconds. The signal shall then change automatically to 1000 cps and held at that value for 30 seconds. In succession, frequencies of 2000, 3000, 4000, and 6000 cps shall be presented to the left earphone. Following these six presentations, the cycle shall be repeated, starting with 500 cps signal directed to the right earphone. At the completion of the test, control shall be taken away from the patient and the signal reduced to zero. It shall be possible to override the machine manually at any time during a test so that a frequency may be skipped or repeated. The frequency shall shift automatically as the recording table is moved manually, without loss of synchronization. The table holding the audiogram chart shall move in a continuous fashion once the actual hearing test is underway. The carriage mechanism shall be driven via a brass gear arrangement. A belt-driven mechanism shall not be acceptable. The audiometer shall have a mid-test delay switch which, when activated, will cause the

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:16Z
Check the source to verify that this is the current version before use.

table to stop at the end of the 6000 cps, left ear presentation. The table movement shall be delayed for 10 ± 3 seconds. During this period, control shall be taken away from the handswitch and the signal strength decreased. At the end of the delay time, the frequency shall change to 500 cps (right ear) and the test shall proceed. The audiometer shall also be equipped with a spring-return switch to suddenly reduce the stimulus level by 10 db. When the switch lever is released, the stimulus shall return to the hearing level corresponding to the stylus position on the audiogram card.

* 3.11 Carrying case. The instrument shall be mounted in a molded ABS plastic carrying case. The audiometer shall be compact and of rugged construction. Suitable accessory storage space shall be provided. The audiometer in the carrying case shall not weigh more than 38 pounds and shall not be greater than 10 x 12-1/2 x 22-1/2 inches overall.

3.12 Audiogram charts. One hundred audiogram charts suitable for recording results of examination shall be furnished with each audiometer. (see 6.2).

3.13 Identification marking. Audiometer shall be provided with a metal nameplate, not less than 0.015 inch thick, bearing the name or registered trademark of the manufacturer, electrical data, manufacturer's model and serial number, contract or purchase order number, Federal stock number, and the letters "U.S.". If the manufacturer is not the contractor, the contractor's name or registered trademark shall also appear on the nameplate. The markings on the nameplate shall be clear and permanent and shall be accomplished by die stamping, embossing, etching or photo process.

3.14 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298 MANUAL, TECHNICAL: Commercial Equipment. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:16Z
Check the source to verify that this is the current version before use.

manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.15 Workmanship. Workmanship shall be first class throughout. Audiometers shall be free from defects which may impair their serviceability or may detract from their appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with inspection level S-1 of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:16Z
Check the source to verify that this is the currentversion before use.

* 4.2.2 For tests. All units shall be tested as specified in 4.4. All defective units shall be rejected.

4.3 Examination. The audiometers shall be examined to determine compliance with all requirements contained in this specification.

* 4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

* 4.4.1 Accuracy of tone frequencies. The accuracy of the tone frequencies generated by the audiometer shall be determined by using a suitably calibrated frequency counter. Frequency error shall not be greater than that specified herein.

4.4.2 Purity of tone. The sound pressure levels of the fundamental frequency and its harmonics shall be measured with the earphone coupled to a National Bureau of Standards Coupler 9-A. Measurements shall be made at all frequencies and at the 60 db hearing-loss setting. The difference between the sound pressure levels of the fundamental and its harmonics shall be reported.

* 4.4.3 Accuracy of attenuator. Measurement of attenuator accuracy shall be made electrically at the input to the earphone terminated in a dummy load of 100 ohms.

* 4.4.4 Accuracy of sound pressure levels. The accuracy of the sound pressure levels at each frequency shall be determined with the earphone coupled to a National Bureau of Standards 9A Coupler. The accuracy of the sound pressure levels at each frequency after a minimum of one hour of operation shall also be determined. The sound pressure levels obtained shall conform to requirements for the initial values.

4.4.5 Stability with respect to variations in supply voltage. The audiometer shall be tested over the full range of supply voltages to determine compliance with requirements. The voltage during test shall be varied from 105 to 125 VAC for 115 volt, 50-60 cycle operation. Tests for compliance with requirements shall be made at a 70 db hearing threshold level setting and at a frequency of 1000 cps.

* 4.4.6 Noise during test.

* 4.4.6.1 Sound from test earphones. The rms weighted sound pressure, due to all frequencies except the signal frequency and its harmonics, shall be determined with the earphone coupled to a National Bureau of Standards 9A Coupler.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

* 4.4.6.2 Sound from second earphone. The sound pressure level in the earphone not being used for test purposes shall be measured to determine compliance with requirements. This test shall be performed by measuring the electrical signal across a dummy load of 100 ohms which simulates the impedance of the earphone.

* 4.4.6.3 Other unwanted sound. A subject having normal hearing shall be located in the normal test position wearing a pair of disconnected earphones. The electrical output of the audiometer shall be absorbed in a 100 ohm resistor while the test is being performed. The audiometer shall be operated through a normal test cycle at a hearing loss setting of 50 db. The subject shall listen for any sound emanating from the audiometer which could furnish a clue that could influence the hearing test results. This test is not intended to apply to sounds from a mechanism such as the hand-operated switch, since the subject knows when he presses this switch.

* 4.4.7 Shock hazard. The audiometer shall be tested to determine if an electrical shock hazard exists. A shock hazard shall be considered to exist at an exposed part if the open circuit potential is more than 25 volts and the current through 1500 ohm resistor is more than 5 milliamperes.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One audiometer, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Secure the card table into position by using 3/4 inch filament tape or by use of a polyurethane foam insert and place the accessory box between the oscillator section and the control panel. One hundred audiogram cards (packaged) shall be placed next to the accessory box and the cover of the audiometer positioned in place and secured.

5.1.2.2 Unit package. Each unit shall be wrapped in 4 mil polyethylene barrier material and the wrap secured. The unit shall be cushioned with two inch formed polystyrene and packaged in accordance with MIL-P-116, method II B. Contact preservative is not required. The exterior container of the method II B pack shall conform to PPP-B-636, type CF, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:16Z
Check the source to verify that this is the current version before use.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each audiometer packaged as specified in 5.1 shall be packed in a snug fitting exterior container conforming to PPP-B-601, overseas type. Closure and strapping shall be as specified in the appendix of the container specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Each audiometer packaged as specified in 5.1 shall not require additional overpacking.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior and exterior containers shall be marked in accordance with MIL-STD-129.

* 6. NOTES

6.1 This specification covers the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-584-3057	Audiometer, Automatic, Portable, 110 volt, 50-60 cycle, AC.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:16Z
Check the source to verify that this is the current version before use.

6.2 Intended use. The audiometer covered by this specification is intended to be used with Stock No. 6515-113-8332 Chart, Recording Instrument, 1000s. This chart is for use with audiometers calibrated to American National Standard (International Standards Organization) reference levels.

6.3 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.4 This specification does not include all types, classes, grades and sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrosions, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-A-36247A(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 86 1426

REPLACES EDITION OF 1 OCT 84, WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.